

Kuramoto-coupled metachronal wave locomotion for pneumatic soft robots: simulation of three lobopodian morphologies

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Abstract

Multi-legged soft robots inspired by Onychophora (velvet worms) and Cambrian lobopodians offer a promising locomotion paradigm for unstructured benthic environments, yet the relationship between body morphology, central pattern generator (CPG) parameters, and locomotion performance remains poorly characterized. We present a MuJoCo-based simulation study of three pneumatic lobopodian robots (HALLU-S, HALLU-F, and HALLU-C) with 7, 9, and 25 leg pairs respectively, each driven by a Kuramoto-coupled phase oscillator CPG producing metachronal waves. We systematically evaluate the effects of CPG frequency, coupling strength, wave direction, phase noise, and incline angle on locomotion metrics including speed, step length, gait quality, and cost of transport. Key findings include: (i) retrograde metachronal waves produce 23–29% higher speeds than direct waves, consistent with biological observations in *Peripatus* (Manton, 1950); (ii) moderate phase noise enhances locomotion speed by up to 119% through a stochastic resonance mechanism in the coupled oscillator network; (iii) inter-leg phase offsets follow the panarthropod $\phi_I = 1/2n$ scaling law, with root-mean-square tracking errors of 2.6° to 4.8° across morphologies. These results establish design principles for CPG-driven soft robots with high leg counts and provide simulation-validated parameters for physical prototype development.

Keywords: central pattern generator, metachronal wave, soft robotics, Kuramoto oscillators, lobopodian, biomimetic locomotion, stochastic resonance

1 Introduction

Locomotion in multi-legged organisms relies on coordinated wave-like patterns of leg activation known as metachronal waves [Manton, 1950, 1952]. Among extant panarthropods, Onychophora (velvet worms) represent a particularly compelling model for soft-bodied multi-legged locomotion: their unjointed lobopod legs, hydrostatic skeleton, and high leg counts

(13–43 pairs) produce robust walking gaits across irregular terrain without the rigid kinematic constraints of arthropod limbs [Oliveira and Mayer, 2019, Nirody et al., 2021].

Central pattern generators (CPGs) provide a biologically grounded framework for controlling rhythmic locomotion in robots [Ijspeert, 2008]. The Kuramoto model of coupled phase oscillators is particularly suited to metachronal wave generation, as it guarantees frequency synchronization while maintaining stable inter-oscillator phase relationships through nearest-neighbor coupling [Kuramoto, 1984, Strogatz, 2000]. Several groups have applied CPG control to hexapod and multi-legged robots [Owaki and Ishiguro, 2017, Yasui et al., 2017], but implementations with high leg counts ($n > 8$) and soft, compliant bodies remain rare.

This paper presents a simulation study of three pneumatic soft robots inspired by lobopodian body plans, ranging from 7 to 25 leg pairs. Each variant is modeled in MuJoCo with physically realistic contact mechanics, compliant body segments, and first-order pneumatic actuator dynamics. A Kuramoto-coupled CPG drives binary solenoid valves that inflate and deflate each leg pair in a metachronal sequence. We conduct five systematic analyses:

1. Frequency sweep: speed, step length, and cost of transport as functions of CPG drive frequency
2. Coupling gain sweep: effect of Kuramoto coupling strength K on gait synchronization
3. Wave direction comparison: direct (anterior-to-posterior) versus retrograde (posterior-to-anterior) metachronal waves
4. Noise robustness: locomotion performance under stochastic phase perturbations
5. Incline performance: locomotion on slopes up to 30°

Results are compared to biological data from *Peripatus* [Manton, 1950] and the panarthropod phase offset scaling law [Nirody et al., 2021].

2 Methods

2.1 Robot morphology

Three robot variants were modeled based on the body plans of extant and extinct lobopodians (table 1). All variants share a common architecture: a segmented flexible body tube (silicone analogue, density 1100 kg m^{-3}) with bilateral pairs of unjointed lobopod legs attached laterally to each segment. Body segments are connected by hinge joints with passive stiffness ($k = 0.08 \text{ N m rad}^{-1}$) and damping ($b = 0.015 \text{ N m s rad}^{-1}$), constrained to $\pm 0.15 \text{ rad}$ in both pitch and yaw. Legs are modeled as capsule geometries (radius 3 mm, length 18 mm for S/F, 15 mm for C) with a single pitch-axis hinge (range $[-0.3, 0.8] \text{ rad}$, $k = 0.02$, $b = 0.005$).

HALLU-C uses a valve zone architecture: 25 leg pairs are grouped into 8 pneumatic zones (3–3–3–3–3–3–3–4 legs per zone), each driven by a single CPG oscillator. This reflects the practical constraint that high-leg-count robots require valve multiplexing to limit pneumatic hardware complexity.

Table 1: Robot variant specifications. Phase offset is π/n_v where n_v is the number of valve groups.

Parameter	HALLU-S	HALLU-F	HALLU-C
Biological model	<i>H. sparsa</i>	<i>H. fortis</i>	<i>Carbotubulus</i>
Leg pairs	7	9	25
Valve groups	7	9	8
Body length (mm)	120	150	300
Segment length (mm)	15.0	15.0	11.5
Segments	8	10	26
Body mass (g)	32.2	40.9	72.4
Phase offset (deg)	25.7	20.0	22.5

2.2 Simulation environment

Simulations were conducted in MuJoCo 3.9.0 [Todorov et al., 2012] using the implicit Euler integrator with a timestep of 2 ms. The ground plane uses a friction model with coefficients $(\mu_s, \mu_d, \mu_r) = (1.0, 0.005, 0.001)$ for the floor and $(1.5, 0.01, 0.01)$ for leg-ground contact (4-dimensional friction cone). Each simulation runs for 20 s with the first 2 s discarded as transient.

2.3 Pneumatic actuator model

Each leg joint is driven by a general actuator modeling a binary pneumatic solenoid valve. The actuator applies force $F = g_a(u - q)$ where $g_a = 0.009$ is the gain, $u \in [0, 1]$ is the control command, and q is the joint position. First-order activation dynamics with time constant $\tau = 60$ ms model the pneumatic delay between valve opening and leg inflation.

2.4 CPG architecture

The CPG consists of n Kuramoto-coupled phase oscillators, one per valve group:

$$\dot{\theta}_i = 2\pi f + K \sum_{j \in \mathcal{N}(i)} \sin(\theta_j - \theta_i \mp \phi_0) + \sigma \xi_i(t) \quad (1)$$

where f is the drive frequency, K is the coupling gain, $\phi_0 = \pi/n$ is the target inter-oscillator phase offset, $\mathcal{N}(i) = \{i - 1, i + 1\}$ is the nearest-neighbor set, and $\xi_i(t)$ is a unit-variance Wiener process with amplitude σ . The sign convention in the coupling term determines wave direction: the minus sign produces an anterior-to-posterior (direct) wave, while the plus sign produces a posterior-to-anterior (retrograde) wave.

Each oscillator drives a binary valve command:

$$u_i = \begin{cases} 1 & \text{if } \sin(\theta_i) > \cos(\pi d) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $d = 0.4$ is the duty cycle (fraction of the stride period during which the valve is open). Default parameters are $f = 0.8$ Hz and $K = 2.0$.

2.5 Metachronal wave coherence

We quantify the quality of the metachronal wave pattern using a modified Kuramoto order parameter:

$$r = \left| \frac{1}{n} \sum_{i=1}^n \exp(j(\theta_i + i\phi_0)) \right| \quad (3)$$

where $r = 1$ indicates a perfect metachronal wave matching the target phase offsets, and $r \rightarrow 0$ indicates desynchronization.

2.6 Performance metrics

Speed. Mean forward speed computed from head sensor displacement during steady-state locomotion.

Step length. Total displacement divided by the number of complete stride cycles, where each cycle corresponds to one full rotation of the first oscillator ($\Delta\theta_1 = 2\pi$).

Straightness. Ratio of net displacement to total path length, where 1.0 indicates a perfectly straight trajectory.

Relative cost of transport. $\text{CoT} = \bar{u}/(mgv)$ where $\bar{u}$ is the mean actuator command magnitude, m is total body mass, g is gravitational acceleration, and v is mean displacement speed. This metric uses actuator commands as a power proxy and is valid only for relative comparison between variants with identical actuator models.

Froude number. $\text{Fr} = v^2/(gL)$ where L is body length, providing a dimensionless speed for comparison with biological organisms.

2.7 Experimental protocol

All simulations are deterministic for a given set of initial conditions. To quantify sensitivity to initial conditions, the frequency sweep (section 3.2) was repeated with three trials per condition, each with independent random perturbations to the initial oscillator phases ($\sigma_0 = 0.05\text{rad} \approx 3^\circ$). The noise robustness sweep (section 3.5) inherently includes stochastic variation and was run with three trials per noise level. The slope analysis uses signed forward displacement to correctly capture backward sliding at steep inclines.

3 Results

3.1 Phase-lag tracking

All three variants achieved stable metachronal wave patterns with tight phase-lag tracking (figure 1). HALLU-S exhibited the best tracking performance with a root-mean-square error

(RMSE) of 2.6° relative to the coupling target of -25.7° (measured mean: -23.2°). HALLU-F showed an RMSE of 4.8° (target -20.0° , measured -15.4°), and HALLU-C achieved 3.7° (target -22.5° , measured -18.9°).

The systematic undershoot of the phase lag target (measured offsets 2–5° less negative than the CPG target) arises from ground reaction forces and body compliance, not from CPG dynamics. The Kuramoto model predicts zero steady-state phase error when all oscillators share the same natural frequency ($\Delta\omega = 0$), which is the case here. The observed deviation therefore reflects the mechanical coupling between the body and the substrate, where ground contact forces during the stance phase resist the phase advance commanded by the CPG.

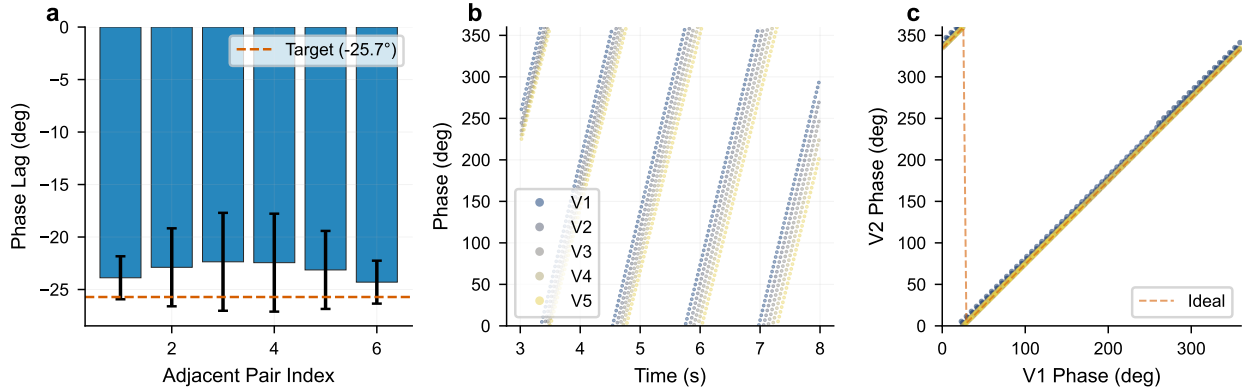


Figure 1: Phase-lag analysis for HALLU-S at $f = 0.8$ Hz, $K = 2.0$. (a) Mean inter-oscillator phase lag per adjacent pair (bars) with standard deviation (error bars) and coupling target (dashed line). (b) Phase trajectories of the first five oscillators showing stable metachronal offset. (c) Phase portrait (θ_1 vs θ_2) with ideal relationship (dashed).

3.2 Frequency sweep

Speed increased monotonically with CPG frequency across all variants (figure 2a), consistent with biological speed–frequency scaling observed in *Peripatus* [Manton, 1950]. At the baseline frequency of 0.8 Hz, HALLU-C (25 leg pairs) achieved the highest speed ($36.2(2) \text{ mm s}^{-1}$), followed by HALLU-F ($32.1(2) \text{ mm s}^{-1}$) and HALLU-S ($30.1(2) \text{ mm s}^{-1}$). Standard deviations across three initial-condition perturbation trials were consistently below 0.6 mm s^{-1} , confirming robustness to initial conditions.

Step length decreased with increasing frequency for HALLU-S and HALLU-F (figure 2b), indicating that the speed increase is driven primarily by stride frequency rather than stride amplitude. Relative cost of transport decreased with frequency across all variants (figure 2d), consistent with the reduced metabolic cost per stride observed in biological multi-legged walkers at higher speeds.

3.3 Coupling gain

Increasing the Kuramoto coupling gain K from 0.5 to 6.0 produced a monotonic decrease in speed (from 43.2 mm s^{-1} to 27.2 mm s^{-1} for HALLU-S) but a monotonic increase in path

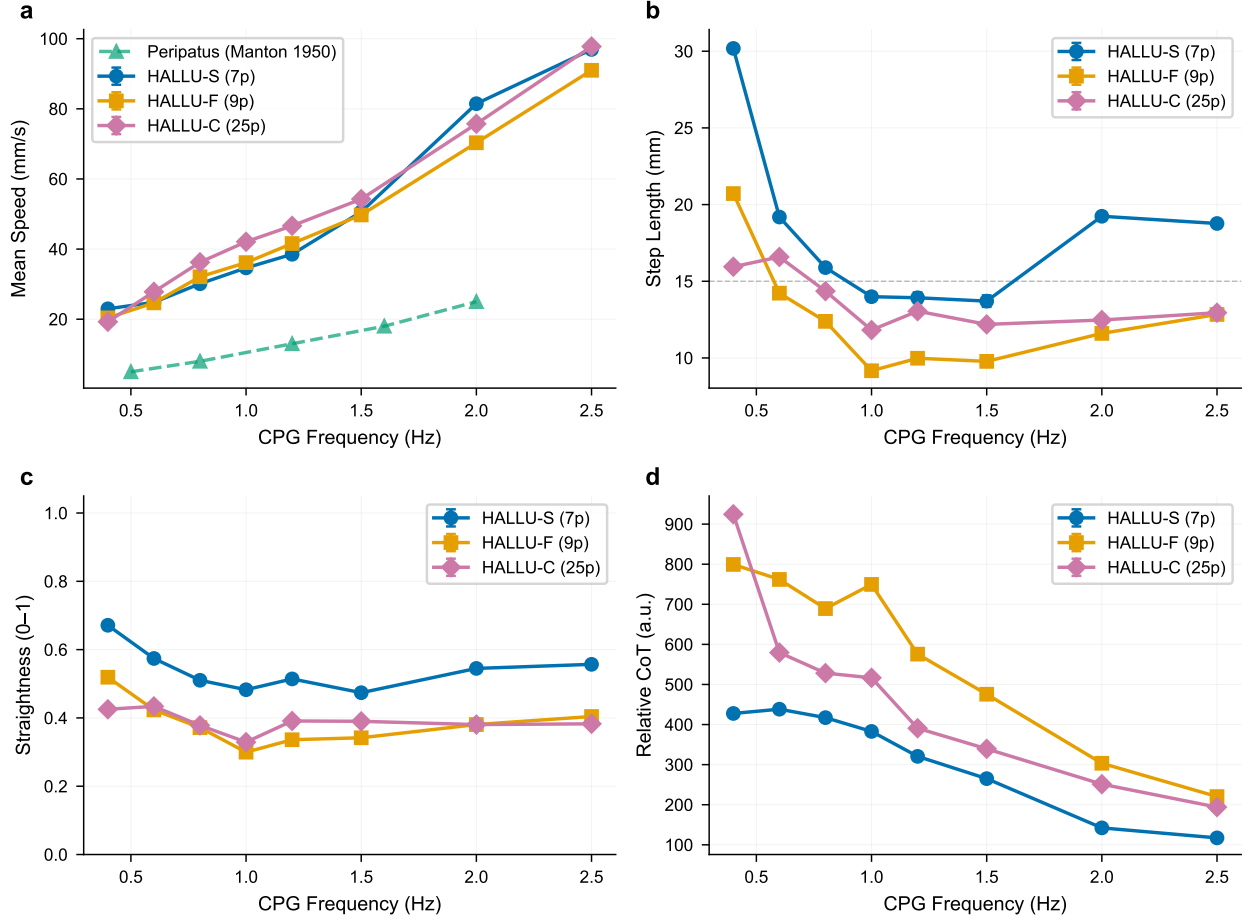


Figure 2: Locomotion metrics across CPG frequency for three morphologies. Error bars show standard deviation over three trials with randomized initial phase perturbation ($\sigma_0 = 0.05$ rad). (a) Speed versus frequency, with *Peripatus* data (Manton, 1950) for biological comparison. (b) Step length. (c) Path straightness. (d) Relative cost of transport.

straightness (0.30 to 0.61) and step length (figure 3). This reflects a fundamental trade-off in coupled oscillator CPGs: stronger coupling enforces tighter phase synchronization, producing more coherent gaits with less lateral drift, but the rigid phase relationships constrain the body dynamics and reduce speed. The design value of $K = 2.0$ represents a compromise between gait quality and locomotion speed.

3.4 Wave direction

Retrograde metachronal waves (posterior-to-anterior activation) produced higher speeds than direct waves (anterior-to-posterior) for HALLU-S (+29%, from 29.7 to 38.4 mm s⁻¹) and HALLU-F (+23%, from 32.5 to 40.0 mm s⁻¹) (figure 4). HALLU-C showed negligible difference between wave directions (< 1%), likely because its valve zone architecture (3–4 legs per zone) averages out the directional effect within each zone.

This result independently reproduces the biological observation by Manton [1950] that

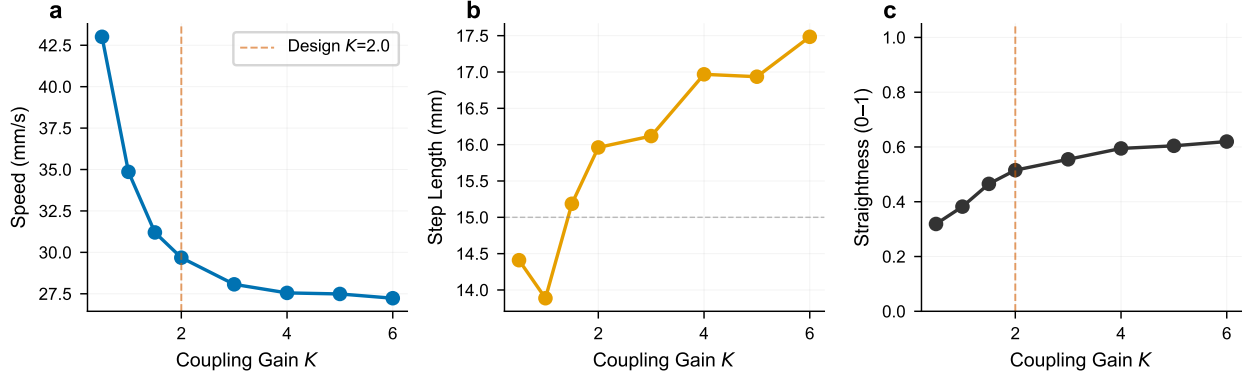


Figure 3: Effect of Kuramoto coupling gain K on HALLU-S locomotion at $f = 0.8$ Hz. (a) Speed decreases with coupling strength. (b) Step length increases above $K \approx 2$. (c) Path straightness improves with stronger phase locking. Dashed line: design value $K = 2.0$.

Peripatus employs retrograde metachronal waves for forward locomotion. The speed advantage of retrograde waves arises because the posterior-to-anterior activation sequence produces ground reaction forces that constructively sum in the forward direction: each leg pushes backward during its power stroke while the leg anterior to it is in recovery, creating a net forward thrust that propagates along the body.

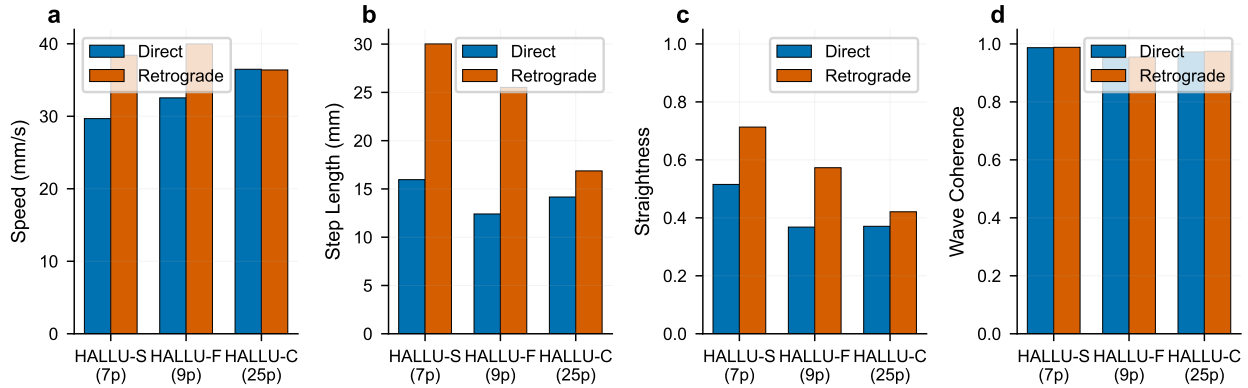


Figure 4: Direct versus retrograde metachronal wave comparison at $f = 0.8$ Hz, $K = 2.0$. (a) Speed, (b) step length, (c) path straightness, (d) wave coherence. Retrograde waves produce 23–29% higher speeds for HALLU-S and HALLU-F.

3.5 Noise robustness and stochastic resonance

Moderate phase noise dramatically enhanced locomotion performance (figure 5). For HALLU-S, speed increased from 29.7 mm s^{-1} at $\sigma = 0$ to a peak of 65.0 mm s^{-1} at $\sigma = 7.0 \text{ rad}/\sqrt{s}$, a 119% increase. Similar peaks were observed for HALLU-F (85% increase) and HALLU-C (85% increase), all peaking near $\sigma = 7.0$. Beyond this optimum, speed degraded as the noise amplitude overwhelmed the coupling and destroyed the metachronal wave pattern (wave coherence $r < 0.3$ for $\sigma > 15$).

This noise-enhanced performance constitutes stochastic resonance in the coupled oscillator network [Gammaitoni et al., 1998]. The mechanism is as follows: at $\sigma = 0$, the deterministic CPG locks into a fixed phase relationship that, due to body-ground mechanical coupling, does not perfectly match the optimal locomotion phase pattern. Moderate noise explores the basin of attraction around the CPG fixed point, transiently visiting phase configurations that interact more favorably with the ground contact dynamics. The result is a higher time-averaged speed despite (or rather because of) the phase variability. At excessive noise, the oscillators desynchronize entirely and coordinated locomotion collapses.

Path straightness showed a complementary pattern: it peaked at low noise, declined moderately through the stochastic resonance regime, and collapsed above $\sigma = 10$. Wave coherence decreased monotonically with noise amplitude, confirming that the speed enhancement does not arise from improved wave quality but from dynamical exploration of the phase space.

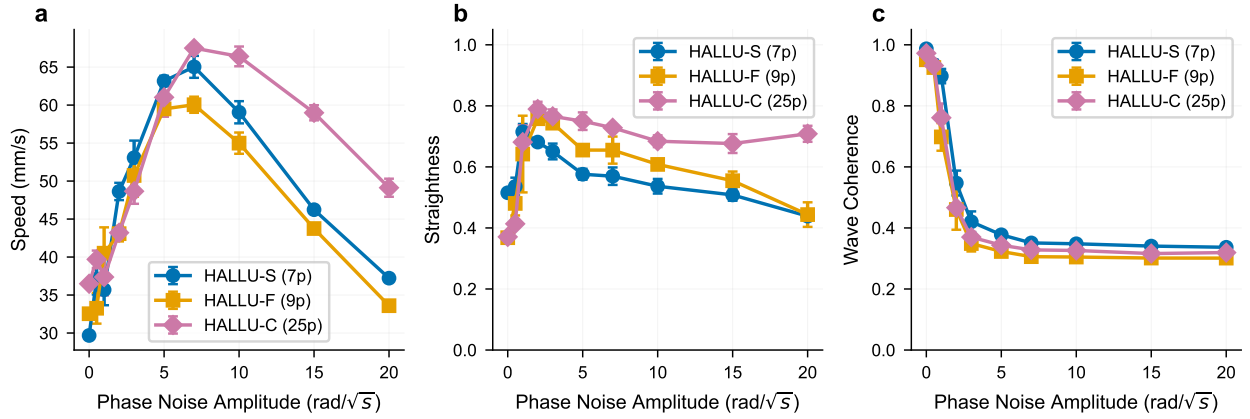


Figure 5: CPG noise robustness sweep. (a) Speed versus phase noise amplitude, showing stochastic resonance peak near $\sigma = 7.0$ rad/√s. (b) Path straightness degrades gradually with noise. (c) Kuramoto wave coherence decreases monotonically. Error bars: standard deviation over three trials.

3.6 Incline performance

Forward speed decreased approximately linearly with slope angle up to 15° , beyond which all variants exhibited net backward sliding (figure 6a). The critical angle (zero net forward speed) was approximately 13° for HALLU-S, 15° for HALLU-F, and 14° for HALLU-C. Cost of transport peaked sharply at angles just below the critical angle (figure 6b), reflecting the high actuator effort required to maintain position against gravity with minimal forward progress. Path straightness was largely unaffected by slope up to 15° , then declined as gravitational forces began to dominate leg thrust (figure 6c).

3.7 Biological comparison

The measured inter-leg phase offsets for all three variants fall on the $\phi_I = 1/2n$ curve (figure 7a), consistent with the half-wavelength metachronal wave produced by the π/n

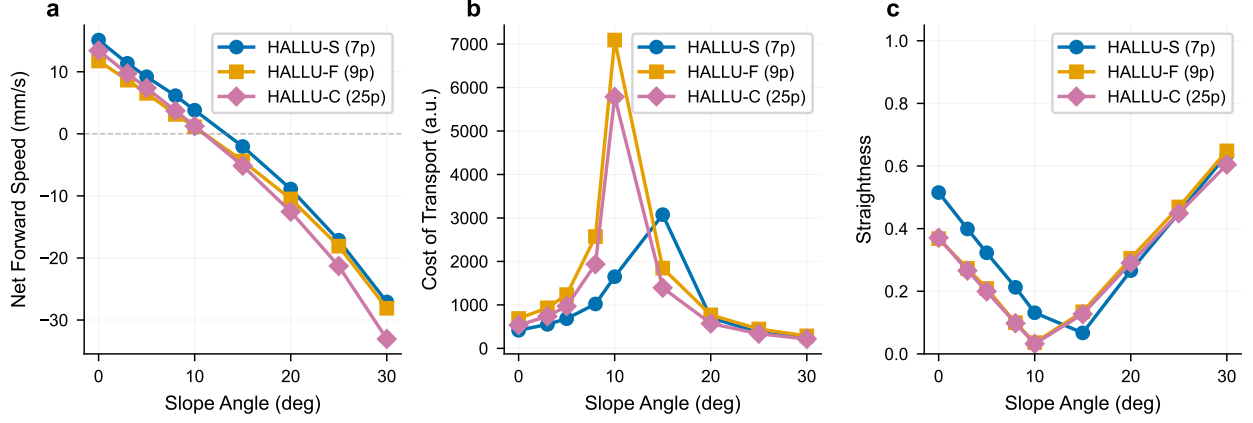


Figure 6: Incline locomotion performance. (a) Net forward speed versus slope angle; negative values indicate backward sliding. (b) Relative cost of transport peaks near the critical angle. (c) Path straightness.

CPG offset. This places our robots below the full-wavelength $\phi_I = 1/n$ scaling observed in biological panarthropods [Nirody et al., 2021], reflecting a design choice rather than a tracking failure: our CPG generates a half-body-length wave whereas biological organisms typically produce full-body-length waves.

Speed–frequency scaling (figure 7b) shows that all three variants exceed *Peripatus* speeds at frequencies above 0.6 Hz, reaching 3–4× biological speeds at 2.5 Hz. When normalized by body length (figure 7c), HALLU-S achieves approximately 0.3 BL/s at 1 Hz, comparable to *Peripatus* at the same stride frequency.

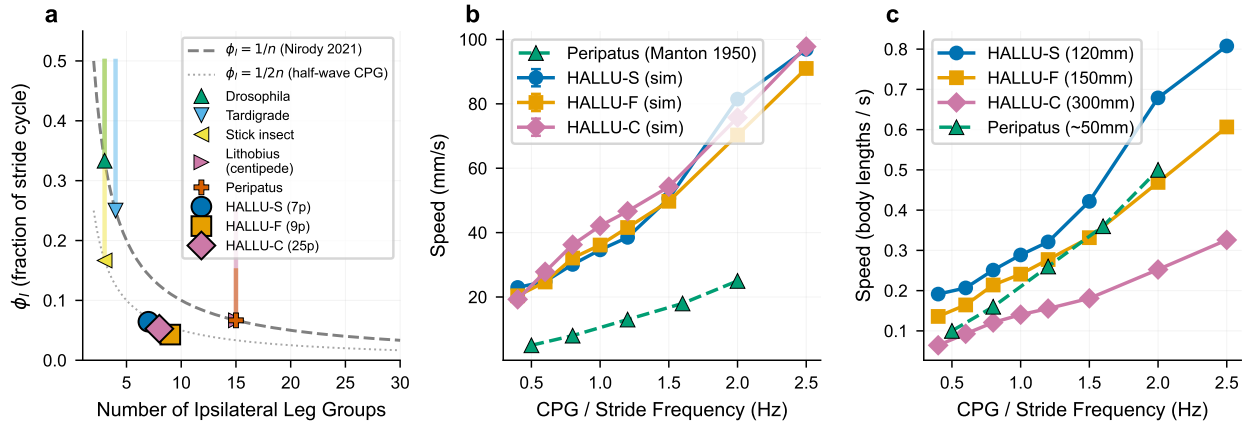


Figure 7: Biological comparison. (a) Phase offset scaling: simulation data (large markers) against the panarthropod $\phi_I = 1/n$ law (Nirody et al., 2021) and biological data points. Vertical bars indicate phase offset range from low to high speed. (b) Speed versus stride frequency with *Peripatus* reference data (Manton, 1950). (c) Size-normalized speed in body lengths per second.

4 Discussion

4.1 Design implications

The results establish several design principles for CPG-driven soft robots with high leg counts:

Retrograde waves. The consistent superiority of retrograde metachronal waves for forward locomotion (section 3.4) suggests that CPG implementations for lobopodian robots should default to posterior-to-anterior wave propagation. This aligns with biological *Onychophora* and provides a 23–29% speed advantage at no additional hardware cost.

Noise tolerance. The stochastic resonance effect (section 3.5) has practical implications for pneumatic soft robots, where valve timing jitter from pressure fluctuations, microcontroller latency, and sensor noise is unavoidable. Our results suggest that moderate timing imprecision may actually *improve* locomotion performance, relaxing the precision requirements for embedded CPG implementations.

Valve zone grouping. The HALLU-C valve zone architecture demonstrates that high-leg-count robots can use valve multiplexing (≤ 8 valve groups for 25 leg pairs) without sacrificing locomotion speed. However, the loss of wave direction sensitivity (section 3.4) suggests that finer-grained valve control would be beneficial if the pneumatic hardware budget permits.

Coupling strength trade-off. The coupling gain K controls a fundamental trade-off between speed and gait quality (section 3.3). For applications requiring straight-line traversal (e.g., pipeline inspection), higher K values are appropriate; for maximum speed (e.g., rapid response), lower K may be preferable.

4.2 Limitations

This study uses simulation exclusively; physical prototype validation is ongoing and will be reported separately. The pneumatic actuator model uses a simplified first-order dynamics approximation that does not capture nonlinear effects such as flow saturation, tube compliance, or pressure-dependent force output. The relative cost of transport uses actuator commands as a power proxy rather than measured mechanical power, and is therefore valid only for relative comparison between variants with identical actuator parameters. Biological reference data from Manton [1950] are approximate values derived from 1950s cinematography and should be interpreted with corresponding uncertainty.

The stochastic resonance finding requires careful interpretation: the speed enhancement occurs at noise levels that degrade wave coherence, suggesting that the mechanism involves dynamical exploration of the phase space rather than improved coordination. Whether this effect persists in physical systems with additional sources of variability (ground irregularity, pneumatic pressure fluctuations) remains an open question for experimental validation.

5 Conclusion

We presented a systematic simulation study of Kuramoto-coupled CPG locomotion for three pneumatic lobopodian soft robots with 7, 9, and 25 leg pairs. The CPG achieves stable metachronal wave patterns with phase-lag tracking errors of 2.6° to 4.8° , retrograde waves provide a 23–29% speed advantage consistent with biological observations, and a stochastic resonance mechanism produces up to 119% speed enhancement under moderate phase noise. These results provide simulation-validated design parameters and control strategies for multi-legged soft robots, with physical prototype fabrication and testing planned as the next phase of this work.

Data availability

Simulation code, MuJoCo models, and all data files are available at <https://github.com/carmenmitchell/hallucigenia-cpg>.

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